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Location awareness and gesture control for an electronically controlled sprinkler system

Abstract

Embodiments of the disclosure relate to automated water sprinkler systems, and more particularly to programmable electronically controlled sprinkler systems which are controllable to affect sprinkler direction and time of watering over a defined coverage pattern. The particular improvements disclosed herein relate to location awareness of the system based on physical location of the unit relative to one or more base platforms, as well as gesture/motion control of the unit for manual operation through the use of an on-board accelerometer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a continuation-in-part of U.S. application Ser. No. 17/577,076, filed Jan. 17, 2022, which is a continuation-in-part of U.S. application Ser. No. 16/439,778 filed Jun. 13, 2019, now U.S. Pat. No. 11,226,054 issued Jan. 18, 2022, which also claims the benefit of U.S. Provisional Patent Application No. 62/815,726, filed Mar. 8, 2019, the entire contents of which are incorporated herein by reference. (2) The present application also claims the benefit of U.S. Provisional Patent Application No. 63/300,089, filed Jan. 17, 2022, the entire contents of which is incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

(1) Embodiments of the invention relate to water sprinkler systems, and more particularly to programmable electronically controlled sprinkler systems which are controllable to affect sprinkler direction and time of watering.

(2) The particular improvements discussed herein relate to location awareness of the system based on physical location of the unit relative to one or more base platforms, as well as gesture/motion control of the unit through the use of an on-board accelerometer.

SUMMARY OF THE INVENTION

(3) A smart sprinkler system in accordance with the present disclosure delivers water from a pressurized water supply to any location within its range via a spray nozzle or sprinkler head that can be rotated in any direction. The distance of the spray field is determined by the water pressure. The term “smart”, in the context of the present system, refers to its ability to be controlled wirelessly using a “smart” device, such as a smart phone or tablet that can run an associated application that communicates with the control electronics of the system. Once configured, the sprinkler functionality is controlled by the device autonomously. This division of labor, the sprinkler device performing the functions of the sprinkler and the phone/tablet providing the Graphical User Interface (GUI), leverages the power of a device that is designed specifically for providing rich and familiar user interfaces, while alleviating the cost and complexity of providing a user interface on the sprinkler.

(4) A defining characteristic of the sprinkler system is that it is able to deliver a small footprint of water to a specific location in a reproducible manner. The sprinkler head rotates about a vertical axis, so the specification of a location is represented in polar coordinates as an angle and a pressure, where the pressure is related to the radial distance from the sprinkler head. The area of water striking the ground can best be envisioned as a narrow rectangle or short line segment radial to the nozzle. The orientation of the rectangle lengthwise along the radius is also intentional, as the primary means of distributing the water in a pattern is by sweeping it about an arc. A sprinkler pattern is created as a series of curves, with each curve being defined as a change of pressure and change of rotation angle. Each pressure represents a distance from the sprinkler head. If the pressure is changed, but the rotation angle is not, then the movement would describe a radial line

segment emanating from the sprinkler head at the center. If the rotation is changed, but the pressure is not, then the movement describes an arc with a radius proportional to the pressure and endpoints corresponding to the starting and ending angles of rotation. If both the pressure and rotation angles change then the movement describes a curve that approximates an average of two arcs at the two pressures with the same rotational endpoints. These movements effectively amount to vectors described in polar coordinates, where the change in pressure represents the radial component and the change in rotation represents the component along an arc. Note that sufficiently small movements may be used to approximate straight lines. As the sprinkler describes a curve over time, a narrow band of spray is produced along the arc component. A user is able to describe a series of points and curves, which together combine into a predefined area, or watering pattern, as the sprinkler sweeps across between each pair of points in sequence.

(5) The electrical components associated with the direct control of the sprinkler system may comprise two low-power DC motors: one for a pressure control valve and the other for a diverter that controls the rotational direction of the spray nozzle. The motors provide input to magnetic couplings. The valve control motor is used to open and close the piloted valve incrementally and the diverter motor is used to rotate an armature to change the flow path in an oscillator which drives the sprinkler head in either direction or may hold it stationary.

(6) The microcontroller unit (MCU) performs all functions related to the control of the sprinkler system using software stored in non-volatile memory within the MCU itself. The MCU also includes storage, both volatile (RAM) and non-volatile, for storing data associated with the running of the device. The MCU also provides a short-range radio used to provide a wireless interface used by the external smart device to configure and control the sprinkler remotely. The firmware defines and implements a command interface to provide these capabilities. Additionally, the MCU provides timing and counting functions that allow the device to control when it starts or stops. It may be configured to, for example, run a user-defined pattern for a specified duration or to be repeated a specified number of times before shutting itself off.

(7) Improvements to the current system include location inputs (RFID tagging) which allows the unit to distinguish between a plurality of different base locations which would each have their own unique program patterns, as well as the provision of hand gesture input (accelerometer) to allow the unit to be controlled in a manual mode when not in a preprogrammed pattern mode or to otherwise react to changes in orientation (e.g. for tip detection).

(8) Accordingly, it can be seen that the present system provides several unique and novel improvement over systems of the prior art, particularly with respect to sealed chamber magnetic couplers which eliminate the need for high friction seals for rotating parts and which also reduce power needs for rotating components within the sealed chambers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments of the present invention can be more readily understood and appreciated by one of ordinary skill in the art from the following descriptions of various embodiments of the invention when read in conjunction with the accompanying drawings in which:

(2) FIG. 1 illustrates an exemplary sprinkler system block diagram in accordance with the present invention;

(3) FIG. 2 illustrates an exemplary configuration of the present sprinkler system;

(4) FIG. 3 illustrates an exemplary spray pattern using an angled nozzle head

(5) FIG. 4 diagrammatically illustrates an area to be watered overlaid with an exemplary programmed watering pattern which can be implemented with software for controlling the present invention;

- (6) FIG. 4A illustrates rotation of the sprinkler head and adjustments of water pressure for distance to transcribe the desired watering pattern;
- (7) FIG. 4B-C diagrammatically illustrate another exemplary programmed watering pattern using a different algorithm where the pressure and rotation are changing concurrently in a vector system with polar coordinates;
- (8) FIG. 5 is a bottom perspective view of an exemplary magnetic coupling and limit switch system illustrating the external drive components;
- (9) FIG. 6 is a side view thereof illustrating both the external drive components and the internal follower components;
- (10) FIG. 7 is a top perspective view thereof illustrating the internal follower arm and mechanical stops;
- (11) FIG. 8 is a top view thereof;
- (12) FIG. 9 is an exploded perspective view thereof illustrating the alignment of the coupling magnets captured in the internal and external coupling components;
- (13) FIGS. 10A and 10B illustrate coupled rotation of the magnets when freely moving and continued rotation and axial translation of the external magnet when the internal magnet is restricted by a mechanical stop;
- (14) FIG. 11 illustrates an exemplary pilot valve which is adapted for variable pressure control with a magnetically actuated lead screw;
- (15) FIG. 12 illustrates a full assembly view of the pilot valve pressure control;
- (16) FIG. 13 is a perspective view thereof shown partially in transparency;
- (17) FIG. 14 is a cross-sectional view thereof taken along line 14-14 of FIG. 12;
- (18) FIG. 15 is an exploded perspective view of the valve control mechanism for the pilot valve;
- (19) FIG. 16 is a plan view of an exemplary oscillator/diverter mechanism in accordance with the present invention;
- (20) FIG. 17 is another plan view thereof shown partially in transparency;
- (21) FIG. 18 is a bottom perspective view thereof showing the external drive components of the magnetic coupling system;
- (22) FIG. 19 is a perspective view of the diverter assembly;
- (23) FIG. 20 is an exploded perspective view of the diverter control magnetic coupling input;
- (24) FIG. 21 is a top view of the diverter control;
- (25) FIG. 22 is an exploded perspective view of the turbine chamber in position above the output of the diverter;
- (26) FIG. 23 is a top view thereof showing the relationship of the diverter armature and the flow ports of the oscillator chamber;
- (27) FIG. 24 is an exploded perspective view of an exemplary embodiment including multiple mounting bases each including a unique RFID for location awareness and sprinkler programming and operation based on the unique location ID; and
- (28) FIG. 25 is an enlarged cross-sectional view taken along line 25-25 of FIG. 24 showing the RFID alignment with an internal RFID reader.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(29) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the device and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present disclosure. Further, in the present disclosure, like-numbered components of the

embodiments generally have similar features, and thus within a particular embodiment each feature of each like-numbered component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape. Further, to the extent that directional terms like top, bottom, up, or down are used, they are not intended to limit the systems, devices, and methods disclosed herein. A person skilled in the art will recognize that these terms are merely relative to the system and device being discussed and are not universal.

(30) Referring generally to FIGS. **1** through **4**, a smart sprinkler system in accordance with the present disclosure is illustrated and generally indicated at **10**. The smart sprinkler system **10** delivers water from a pressurized supply **12** to any location within its range via a spray nozzle or sprinkler head **14** that can be rotated in any direction. The spray nozzle **14** includes at least one, but preferably a plurality of, angled orifices **16** which create an elongated, somewhat narrow spray field **18** as generally illustrated in FIG. **3**. The distance “D” of the spray field **18** is determined by the water pressure. The term “smart”, in the context of the present system **10**, refers to its ability to be controlled (wired or wirelessly) using a “smart” device **20**, such as a smart phone or tablet that can run an associated application that communicates with the control electronics **22** of the system **10**. The present system is implemented in a wireless configuration, and in particular, the wireless interface implemented in this exemplary system is a Bluetooth Low Energy (BLE) interface which is common to the category of personal devices known as smart devices (e.g. mobile phones, tablets).

(31) It should be understood that other wired and wireless interfaces and standards could also be implemented with the same functionality. Indeed, the smart sprinkler system **10** requires a smart device **20** to control and configure the sprinkler system **10** using an associated application as noted above. Any electronic interface that is capable of supporting commands is a viable possibility. Once configured, much of the sprinkler functionality is controlled by the device autonomously. This division of labor, the sprinkler device **10** performing the functions of the sprinkler and the/phone/tablet device **20** providing the GUI, leverages the power of a device that is designed specifically for providing rich and familiar user interfaces, while alleviating the cost and complexity of providing a user interface on the sprinkler.

(32) A defining characteristic of the sprinkler system **10** is that it is able to deliver a small footprint **18** of water to a specific location in a reproducible manner. There are several possible methods for accomplishing this goal.

(33) The sprinkler head **14** rotates about a vertical axis, so the specification of a location can be represented in polar coordinates as an angle and a pressure, where the pressure is related to the radial distance D from the sprinkler head. The footprint of water striking the ground can best be envisioned as a narrow rectangle or short line segment radial to the nozzle **14**. The footprint **18** is by design and represents a balance between having a fine resolution for placing water and avoiding too high a density of water striking the ground at a single location. The orientation of the rectangle lengthwise along the radius is also intentional, as the primary means of distributing the water in a pattern is by sweeping it over a curve.

(34) In a first exemplary method, a sprinkler pattern is created as a series of arcs **24a**, **24b**, **24c** (FIGS. **4** and **4A**), with each arc being defined as a pressure and two angles. The pressure represents the radius of the arc and the two angles represent the left and right ends or stops **26A**, **26b** of the associated arc. As the sprinkler describes the arc **24** over time, a narrow band of spray **18** is produced. A user is able to describe a series of arbitrary concentric arcs **24**, which together combine into an arbitrary defined area as the sprinkler sweeps across each arc in sequence.

(35) In a second exemplary method, the curve is defined by two points P1, P2 with a change in

pressure and change in rotation angle, relative to the sprinkler head **S**, which defines the center of a circle. The curve is, essentially a vector **v** in polar coordinates with a radial component **r** and an arc component **a** (FIG. **4C**). A sprinkler, or watering, pattern is created as a sequence of such vectors **24a . . . 24n** (FIG. **4B**). As the sprinkler describes each curve **24** over time, a narrow band of spray **18** is produced. In describing a sequence of curves, a path is followed, which in totality deposits a volume of water on an area. A user is able to describe an arbitrary sequence of these curves **24** by defining the points **P1, P2, Pn**. A schematic illustration of an area to be watered overlaid with such a sequence **24a . . .** is illustrated in FIG. **4B**.

(36) It is also a defining characteristic of the sprinkler system **10** that it is able to operate with only a single connection to a water source **12** and in some embodiments does not require a connection to external electric power. This allows the present sprinkler system **10** to be a direct replacement for a typical lawn sprinkler, which is mechanically driven entirely by the energy provided by the supply pressure of the water source. Unlike a mechanical lawn sprinkler, which only provides one pressure (that of the supply) and one set of stops and can, therefore, only supply one pattern that is either roughly rectangular or circular (depending on the type), the present sprinkler system can create arbitrarily shaped patterns. It achieves this by using a circular lawn sprinkler mechanism to drive the rotation of the sprinkler head **14**, but with an electric motor (oscillator/diverter mechanism **300**—described below) to actuate the mechanism that controls the direction of the rotation.

Additionally, present the sprinkler system **10** explicitly controls the water pressure delivered to the spray nozzle, up to the maximum of the supply pressure (i.e. there is no pump to add pressure above that of the supply). The system achieves this using an adjustable piloted valve (pilot valve assembly **200**), which maximizes the use of the supplied pressure to affect changes to the valve. In short, the mechanism is designed to use as little energy as practicable.

(37) In the present wireless configuration, required electrical energy is harvested from two sources: a hydro generator **28** in line with the water flow between the pressurized supply **12** and the nozzle **14** (input and output, respectively) and a solar panel **30**. Both sources provide DC electricity used to power the electronics **22**, MCU/memory **32**, wireless radio **34**, motor control **36** and sensors **38** that comprise the control system.

(38) The hydro generator **28** provides energy while the sprinkler is active (i.e. water is flowing through it) and the solar panel **30** captures energy from the sun, regardless of whether the sprinkler is active. Since the sprinkler is always consuming energy, the power system includes a rechargeable battery **40** and a battery charge controller **42**. Thus, the two power sources **28, 30** are used to put energy into the battery **40**, even as the device consumes energy. The energy system is balanced, such that it requires no additional power source under normal use. There is, however, an electrical connection (not shown) for attaching to an external charger for expediting an initial charge before first use or after storage.

(39) In a wired system, the rechargeable battery **40**, charge controller **42**, hydro generator **28** and solar panel **30** could be eliminated to reduce complexity and cost.

(40) The electrical components associated with the direct control of the sprinkler system **10** may comprise two low-power DC motors: one motor **202** for a piloted pressure control valve system **200** (See FIGS. **11-15**) and the other motor **302** for a diverter system **300** (See FIGS. **16-23**) that controls the rotational direction of the spray nozzle **14**. The motors **202, 302** provide the input to magnetic couplings **204, 304** (further described below). The speed, duration and direction of the motors **202, 302** are controlled by algorithms running on a Microcontroller Unit (MCU) **32**. The valve motor **202** is used to open and close a piloted valve system **200** incrementally and the diverter motor **302** is used to rotate an armature of the diverter assembly **300** to change the flow path in an oscillator which drives the sprinkler head **14** in either direction or may hold it stationary.

(41) In order to provide closed loop control over the system, there are two sensors: a pressure sensor **400** and a magnetic rotational sensor **500**. The pressure sensor **400** is embedded in the oscillator chamber of the oscillator/diverter assembly **300** and measures the water pressure

delivered to the nozzle **14**. When the desired pressure setpoint has been configured, the pressure sensor **400** supplies input to the algorithm, which opens and closes the pressure control valve **200** to maintain the nozzle pressure about the setpoint within a hysteresis range. This allows a pressure to be maintained without calibration of the supply pressure and as the supply pressure changes over time. Note that the output pressure is limited at an upper end by the native supply pressure (i.e. there is no mechanism for increasing the pressure beyond the supply pressure).

(42) It should also be noted that additional environmental sensors **39**, or smart sensors may also be deployed with and communicate with the present control system **32** (wired or wirelessly) to provide additional operational input. Sensors **39** may comprise temperature sensors, atmospheric pressure sensors, light sensors, rain sensors, moisture sensors, infra-red heat sensors etc. to provide additional inputs to control or modify run days, run times, or run locations as configured.

(43) The magnetic rotational sensor **500** is external to the oscillator chamber of the oscillator/diverter assembly **300** and in line with a magnet that is mechanically held to the rotational axis of the sprinkler head **14** or indirectly in correspondence with the sprinkler head **14**. The magnetic field is diametric to the axis of rotation, which allows the sensor **500** to determine the angle of rotation of the sprinkler head **14** within a resolution less than one degree. This allows the algorithm to determine when the sprinkler head **14** has rotated to a desired angle and to then change the direction of rotation or stop the rotation using the motor **302** associated with the diverter armature.

(44) The MCU **32** performs all functions related to the control of the sprinkler system **10** using software stored in non-volatile memory (firmware) within the MCU itself. The MCU **32** also includes storage, both volatile (RAM) and non-volatile, for storing data associated with the running of the device (e.g. the data defining a user-defined pattern). The MCU **32** also provides the radio **34** (Bluetooth) used to provide a wireless interface used by the external smart device **20** to configure and control the sprinkler remotely. The firmware defines and implements a command interface to provide these capabilities. Additionally, the MCU **32** provides timing and counting functions that allow the device to control when it starts or stops. It may be configured to, for example, repeat a user-defined pattern (See FIG. **4**) for a specified duration or a specified number of times before shutting itself off.

(45) FIG. **1** shows a block diagram of the components of the sprinkler **10** that enable all of the capabilities described above. The ultimate purpose of the system **10** is to deliver water from a supply to desired locations. FIG. **1** illustrates the path of the water flow (thicker arrows “W”) through the system as it controls the water and extracts energy from the water to power the system. FIG. **1** also illustrates the path of energy (thinner arrows “E”) as it is generated from the hydro generator **28** and solar panel **30**, stored in the battery **40** and utilized. The system is also responsible for control, so the diagram shows the path of the control and data signals (lines “C”). Finally, the system offloads the responsibility for providing a user interface to the external smart device **20** and the wireless control interface for communication with the external smart device **20** is also illustrated.

(46) Referring to FIG. **2**, the water supply flows, first, into the pressure control valve assembly **200**. The pressure control valve **200** is capable of shutting off the water supply entirely as well as providing a desired set water pressure from zero up to roughly the maximum limit of the native supply pressure. The water then flows into the hydro generator **28**. A minimum amount of water pressure is required before the turbine inside the generator **28** will spin. Once it begins to spin, the output energy is relative to the speed of rotation or the output pressure of the control valve assembly **200**. The water flow exits the hydro generator **28** and enters the oscillator/diverter assembly **300**, which determines the direction of rotation of the sprinkler head **14**.

(47) The oscillator/diverter **300** achieves this by directing water through ports that lead to an oscillator turbine. Generally, each port directs water to one side or the other of the turbine, each port corresponding to one direction of rotation. The rotating turbine provides the mechanical

energy to turn the sprinkler head **14**. A diverter chamber also provides a direct path to the sprinkler head through a pressure relief valve. This allows excess water pressure to bypass the oscillator, which limits the maximum speed of rotation of the sprinkler head **14**.

(48) The electrical energy of the system includes rechargeable battery **40**, allowing the sprinkler system **10** to be run over a wide range of circumstances. The battery **40** allows the energy to be budgeted, so that the energy generated does not have to be explicitly associated with energy consumption of specific components. Rather, generated energy is added to the battery **40** and energy consumed comes from the battery **40**. This is all managed by the battery charge controller **42**, which also performs voltage regulation. The battery charge controller **42** directs energy from the hydro generator **28**, which is in the range of 0-5V, and the solar panel **30**, which is in the range of 0-6V, into the rechargeable battery **40** (See FIG. 1). The variability of voltage stems from the fact that the generator **28** is not always running, and its RPMs are relative to the output water pressure. The solar panel **30** only produces energy during the day and its output is relative to the amount of direct sunlight. The energy consumed depends on whether the sprinkler is running. When it is running, the electronics associated with the motors **202**, **302** and the sensors **400**, **500** require 5V and represent the bulk of the energy consumption. The motor control circuit **36** and the sensors **400**, **500** are powered only while the sprinkler is active. The motors **202**, **302**, which are the greatest consumers of energy by a factor of 10 are only driven intermittently and for relatively short durations. The MCU **32**, which includes the radio **34**, is always consuming some amount of energy, even when the sprinkler is off. Running at 3.3V, the energy consumption is less at its maximum, when the radio **34** is actively linked to the external smart device **20**, than the electronics. The energy consumption is smaller when the system **10** is running autonomously versus when the system **10** is being remotely controlled. The control signals that trigger the motor driver and sense the output of the pressure and rotation sensors are components of this low voltage/power consumption.

(49) Turning now to FIGS. 5-10, an exemplary magnetic coupling and switch assembly for use with a sealed chamber system is generally indicated at **100**. The assembly **100** is generally used to transfer a rotational mechanical input force that is external to a sealed chamber **102** to a rotational mechanism that is internal to the chamber **102** which may be filled with fluid **104** under pressure. The exemplary embodiment shown in FIGS. 5-10 is a universal configuration that could be used in any sealed chamber system to translate external motion to an internal component and to provide a linear (axial) switching actuator external to the chamber **102**. These general magnetic coupling and actuator principles are used in conjunction with two separate systems described with the present system hereinafter. First, for rotation of the lead screw of a variable pressure pilot valve system generally indicated **200** (FIGS. 11-15). Second, for rotation of an oscillator/diverter lever in a water driven turbine rotation system **300** for the spray nozzle (FIGS. 16-23).

(50) The magnetic coupling assemblies have the following characteristics:

(51) The internal chamber **102** is hermetically sealed. There are no mechanical components penetrating the chamber which would require hydraulic seals.

(52) The connection between the external and internal members introduce minimal friction, due to the elimination of seals and due to the force vectors associated with the magnetic fields.

(53) The magnetic coupling uniquely provides the ability for an external sensor to detect when the internal mechanism has reached a hard stop.

(54) The magnetic coupling also inherently functions as a safety clutch, allowing the coupling to disengage without damage or wear.

(55) FIGS. 5-8 illustrate the components that are both external to the chamber **102** and internal to the chamber. The boundary between internal and external is defined by the wall **106**. The inside of the chamber **102** being above the wall **106** and further defined by walls **108** illustrated in broken line in FIG. 6. In the drawing figures, the external components are below the boundary wall **106**. The active external components are mounted on a fixed stand **110**, which maintains the position

and orientation of the external components relative to each other and to the internal components. The exemplary mechanism is driven by a reversible motor **112**. A motor gear **114** is mounted to the motor shaft (not shown). The motor gear **114** engages with and drives a magnetic coupling gear **116**. The magnetic coupling gear **116** has a square hole **118** traversing it from top to bottom (See FIG. **9**). In this hole, is an external square magnet **120** (FIG. **9**) abutting the chamber boundary wall **106** and a magnet adapter **122** (also in FIG. **9**) that couples the magnet **120** to a limit switch **124**. Note that a bracket holding the limit switch **124** is not shown. The magnet **120** and the magnet adapter **122**, being square are not able to rotate freely within the magnetic coupling gear **116**, so that the rotation of the magnetic coupling gear **116** is transmitted to the magnet **120** and the magnet adapter **122**, making them also driven by the coupling gear **116**. The magnet **120** and the magnet adaptor **122** can move freely in the direction perpendicular to the chamber boundary wall **106** (axially along the axis of rotation of the magnetic coupling gear **116**). This axial motion of the magnet **120** causes the limit switch **124** to actuate, closing and opening as the magnet **120** and magnet adaptor **122** move away from and toward the chamber boundary wall **106**, respectively. (56) FIGS. **7-9** best illustrate the components that are internal to the chamber **102**. The chamber boundary wall **106**, as shown, incorporates a well **126** that holds a rotating armature **128** in place. The armature **128** sits inside a bushing **130** inside the well to reduce friction. The well **126** also reduces the thickness of the boundary wall **106**. The armature **128** includes a square socket **132** that captures another internal square magnet **134** (best seen in FIG. **9**). Opposite the well, an axle **136** supports the rotation of the armature **128** on an axis perpendicular to the chamber wall **106**. The axle **136** is captured by a fixed mount **138** and a washer **140**. The mount **138** incorporates stops **142a**, **142b** that restrict the rotation of the armature **128** (see FIG. **8**).

(57) FIG. **9** presents a simplified exploded view that shows the alignment of the rotational components of the assembly; both internal and external to the chamber **102**. The components below the boundary wall **106** are external to the chamber **102** and those above are internal. The boundary wall **106** is shown in cross section and shows the reduced thickness of the boundary wall at the bottom of the well **126** in which the internal components are seated. The motor gear **114** is shown for reference without the motor **112** that drives it.

(58) Magnetic coupling is effected by the external magnet **120** and the internal magnet **134**. The poles of the magnets **120**, **134** are parallel to the boundary wall **106**. In the diagram, they are shown in phase; the north and south poles of the external magnet **120** are aligned with the south and north poles, respectively, of the internal magnet **134**. This supplies the maximum attractive force normal to the boundary wall **106** with a net torque of zero about the center line (axis).

(59) The external magnet **120** is captured by the square hole **118** in the magnet coupling gear **116**. As the magnetic coupling gear **116** is driven by the motor gear **114**, the magnet **120** is correspondingly rotated about the center line. The magnet adapter **122** is also located in the square hole **118**, adjacent to the magnet **120** and between the magnet **120** and the limit switch **124**. It is also square and rotates in correspondence to the magnetic coupling gear **116** and the magnet **120**. It serves to mate the square magnet **120** to the round button on the limit switch **124**. The magnet **120** and magnet adaptor **122**, while rotationally restricted, are free to move normal to the boundary wall **106**.

(60) Inside the chamber **102** (above the wall **106**), the internal magnet **134** is captured by the square socket **132** in the rotating armature **128**. The armature **128** is seated in the well **126** of the boundary wall **106** inside bushing **130**, which allows the armature **128** to rotate freely, in correspondence with the rotation of the internal magnet **134**. The armature **128** is secured at the surface opposite the boundary by the axle **136** and washer **140** held by the fixed mount **138**. This arrangement allows the armature **128** to rotate but restricts the movement of the armature **128** normal to the boundary wall **106**.

(61) Turning to FIGS. **10A-10B**, the behavior of the magnetic coupling is driven by the motor **112**, which rotates the motor gear **114**. The motor gear **114**, in turn, rotates the external magnet coupling

gear **116**, which rotates the external magnet **120**. As the external magnet **120** rotates, it becomes out of phase with the internal magnet **134**. The force vector associated with the magnets **120**, **134** is proportionally skewed from normal and develops a rotational component. The resultant torque increases as the phase angle increases. At some point, the torque increases enough to rotate the internal magnet **134** (See FIG. **10A**). The internal magnet **134** rotation drives the armature **128**, accordingly.

(62) Rotation of the armature **128** is restricted by stops **142a**, **142b** (see FIG. **8**). Once the coupling is rotated to the point of contacting a stop **142**, the armature **128** can no longer rotate. At this point, the external magnet **120** continues to be driven, increasing the phase angle between the two magnets **120**, **134**. When the phase angle increases to 90 degrees, the force vector becomes completely rotational; there is no force normal to the boundary. As the angle increases beyond 90 degrees, the force normal to the boundary reverses, becoming repulsive. At some point, the repulsive force become large enough to push the external magnet **120** away from the boundary wall **106** (see FIG. **10B**). The repulsive force is transmitted through the magnet adapter **122**. At some point, the repulsive force increases enough to force axial movement of the magnet **120** and magnet adapter **122** and activate the spring-loaded limit switch **124**. The activation of the limit switch **124** is detected electronically, and the motor **112** is stopped.

(63) The motor **112** is then reversed via the electronics, which decreases the phase angle. As the phase angle decreases, the repulsive force decreases until it is nullified at the 90 degree phase angle. As the motor **112** (and external magnet **120**) continues to rotate then the force becomes attractive and increases to its maximum at the zero degree phase angle. At some point, the external magnet **120** moves back toward the boundary wall **106** and releases the limit switch **124**. The motor **112** is stopped again, once the limit switch **124** is deactivated.

(64) This reverse motion that is terminated with the release of the limit switch **124** is called the back-off period. When rotating the coupling towards one of the stops **142**, the external magnet **120** is intentionally over-rotated after the armature **128** is physically stopped. The over-rotation causes an increase in the phase angle. Since the limit switch **124** cannot be activated, except via a repulsive force between the magnets **120**, **134**, the limit switch **124** is not engaged until the phase angle exceeds 90 degrees. The limit switch **124** activation marks the beginning of the back-off period, when the motor **112** is reversed. The limit switch **124** cannot be disengaged until the repulsive force is removed and the magnetic force transitions from repulsion to attraction. Therefore, the limit switch **124** is released when the phase angle is reduced to less than 90 degrees, which marks the end of the back-off period. At the end of the back-off period, the phase angle is such that the magnetic force vector holds the armature **128** against the boundary and against the stop **142**. Note that this assumes that the motor **112** and motor gear **114** are locked in position.

(65) In alternative arrangements, the limit switch **124**, which operates to sense armature **128** position, could be replaced with a solid state sensor, such as optical or magnetic sensors. For example, for an optical sensor system the chamber boundary wall **106** separating the magnets **120**, **134** could be optically clear. There could be optical sensors (2) (not shown) located strategically at the hard stops inside the chamber **102**. The sensor could then pick up on the armature **128** inside being in position at the hard stop **142**. From a power resource view, even though these sensors are, themselves, consuming energy, it would actually be a win energy-wise because the limit switch **124** requires the motor **112** to be over-rotated to activate the limit switch and then the motor must be reversed to perform the back-off movement. The extra back and forth movement represents an additional 180+ degree movement that would not be required if using such an optical solid state sensor. Since the motor itself is a much larger consumer of energy, the extra movement is costly. Also, the sensors can be turned off until they are actually needed (i.e. just before the motor movement). On balance, using a solid state sensor may be a significant improvement energy-wise and time-wise, since the motor over rotation movement does take time.

(66) As noted, the generic magnetic coupling system **100** described above is useful in any sealed

chamber system where there is a need to reduce seal and friction between moving parts. Further exemplary coupling systems are described below in connection with the piloted valve system **200** and the oscillator/diverter assembly **300**.

(67) Referring now to FIGS. **11-16**, there is shown and described an exemplary pressure control valve assembly **200** controlled by magnetic coupling assembly **204** having the operational characteristics as generally described hereinabove.

(68) The pressure control valve **200** adjusts the water pressure delivered to the spray nozzle **14**. The distance of the spray is proportional to the pressure. In the exemplary system, the valve **200** is controlled by a motor **202**, making it capable of being controlled algorithmically by microprocessor **22**. The system **200** incorporates pressure sensor **400** located adjacent the nozzle **14**, which allows the system to operate as a closed loop with respect to water pressure.

(69) The basis of the control valve **200** is a pilot valve **206** which is used to control the water pressure delivered to the sprinkler head **14**. The pilot valve **206** may comprise three chambers: the input chamber **208**, the output chamber **210** and the control chamber **212** (see FIGS. **13-14**), attached to the top of a valve seat **214**. The input chamber **208** is supplied with water through an input port **216**. The pressure in the input chamber **208** is always the maximum pressure, which is referred to as the supply pressure. The output chamber **210** is vented to the atmosphere via an outlet **218** and, ultimately, to the spray nozzle **14**. Atmospheric pressure is considered the zero pressure.

(70) When the valve **200** is closed, the pressure in the output chamber **210** is zero.

(71) When the valve **200** is open, there is, typically, back pressure due to the relatively narrow orifice of the nozzle **14**. Therefore, the output pressure may be greater than zero, but is always less than the supply pressure.

(72) The primary flow of water flows between the input chamber **208** and the output chamber **210** directly when the valve is open. A secondary flow path is through the control chamber **212**, which is separated from the input/output chambers **208**, **210** by a flexible diaphragm **220** that incorporates into it a rigid stabilizer **222**. The stabilizer **222** is connected to the diaphragm **220** by a plurality of circumferentially spaced posts **223** (only one visible) which are press fit through corresponding holes **225** in the diaphragm **220**. Four points of connection provide higher rigidity to the diaphragm **220**, prevent vibrational instability caused by water flow and allow the valve to stabilize in the desired position more quickly.

(73) Water flows into the control chamber **212** through an input pilot hole **224**, which is always open. Pilot hole **224** extends through one of the connection posts **223**. The water flows from the control chamber **212** to the output chamber **210** via an output pilot hole **226**, which may be opened or closed by a plug **228**.

(74) The benefit of the pilot valve arrangement is that a very small amount of energy is necessary to open or close it. The control input is a matter of opening or closing the output pilot hole **226** using the plug **228**. The amount of energy involved is very small because the force necessary is a product of the very small area of the output pilot hole **226** and the pressure differential between the control chamber **212** and the output chamber **210**. Once the pilot hole **226** is opened or closed, the primary flow of water between the input chamber **208** and output chamber **210** is affected by the position of the diaphragm **220**, which is a function of the force differential on the two sides of the diaphragm **220**. On one side of the diaphragm **220**, the force is a product of the area of the diaphragm **220** and the pressure in the control chamber **212**. On the other side, the total force is the sum of the force on the area adjacent to the input chamber **208** and the force adjacent to the output chamber **210**. Note that the area adjacent to the input chamber **208** is significantly larger than the output area.

(75) When the plug **228** is blocking the output pilot hole **226**, the input pilot hole **224** causes the input chamber **208** and control chamber **212** to equalize, so the forces on both sides of the diaphragm **220** corresponding to the input area are equal. Since the output pressure is always less than the supply pressure, the area adjacent to the output chamber **210** is less than the corresponding

area in the control chamber **212**. This differential causes the diaphragm **220** to press toward the output port **218**, restricting water flow and, ultimately, closing and sealing the output port **218**. When the plug **228** is not blocking the output pilot hole **226**, the pressure in the control chamber **212** equalizes with the output chamber **210**. In this state, the force on both sides of the diaphragm **220** adjacent to the output area is equal. Since the control chamber pressure is less than the input chamber pressure, the force on the area adjacent to the input chamber **208** is larger on the input chamber side and the diaphragm **220** is pushed away from the output port **218**, allowing more water to flow. Note that when the diaphragm **220** is fully open, there is necessarily a pressure differential between the supply pressure and the control/output pressure because it is the differential that is holding the diaphragm open. That means there is a small loss of pressure when using this type of valve.

(76) FIGS. **12-14** show the full valve assembly **200** with the pilot valve mechanism at the bottom with its input port **216**. The basic function of the pilot valve chambers **208**, **210**, **212** and diaphragm **220** is conventional in the art. However, the method of controlling the valve **200** is unique. The control chamber **212** is comprised of two parts **212a**, **212b**, as shown, and encompasses the control components of the valve. The external drive for movement of the plug **228** is supplied via a magnetic coupling **100** generally as described hereabove, the external parts of which are shown and include motor **202**, stand **230**, drive gear **232**, magnetic coupling gear **234**, external magnet **236**, magnet adapter **238** and limit switch **240**. The internal parts of the magnetic coupling **100** drive a control mechanism **242** that results in the plug **228** moving normal to the output port of the diaphragm **220**.

(77) FIG. **15** shows the control mechanism **242** internal to the control chamber **212**.

(78) The mechanism **242** is driven rotationally by the external magnet **236** (FIGS. **12-14**), causing an internal magnet **244** to rotate, accordingly. The internal magnet **244** is captured in a square socket at the top of a magnet adapter **246**. The magnet adapter **246** rotates within bushings **248** and **250** at top and bottom respectively inside a cylindrical void in the control chamber housing **212** (FIG. **14**). The magnet adapter **246** incorporates slots **252** that trap a control arm **254** that is embedded radially in a threaded leadscrew **256**. The leadscrew **256** engages a nut **258** that is captured by the control chamber housing **212**. Two bushings **260a**, **260b** provide the bearing surface between the rotating magnet adapter **246** and the stationary nut **256**. The bottom of the leadscrew **256** incorporates a flange **262** around which the rubber plug **228** is molded. A leadscrew guide **264** ensures the plug **228** is centered on the output pilot hole **226**. It also serves to lock the diaphragm **220** in place in the pilot valve mechanism. Note that it is stationary.

(79) The pilot valve control mechanism **200** operates in the same manner as the magnetic coupling mechanism **100** described hereinabove. The external motor **202** drives the external magnetic coupling gear **234** and external magnet **236**, which then drives the internal magnet **244**. The square magnet **244**, being trapped in a square socket rotates the magnet adapter **246**, which rotates the leadscrew **256** via the control arm **254**. As the leadscrew **256** turns in the trapped nut **258**, it moves up and down relative to its rotational axis. This causes the plug **228** to unblock and block the output pilot hole **226**, accordingly. When the pilot hole **226** is unblocked, the diaphragm **220** rises and increases water flow through the valve **200**. When the diaphragm **220** rises and contacts the plug **228**, the pilot hole **226** is blocked, and the diaphragm **220** is pushed back down by the control chamber pressure and the water flow decreases. As equilibrium is reached, the resulting behavior is that the diaphragm **220** follows and is positioned by the control mechanism **242**. This allows the external motor **202** of the magnetic coupling arrangement **204** to effectively control the water flow through the valve **200** and, ultimately, the water pressure delivered to the sprinkler nozzle **14**.

(80) The pressure control valve **200** uses the magnetic coupling **204** for the control input and includes limit switch **240** as also described above. The limit switch **240** is not used to control the variable pressure, but it is used to ensure that the valve **200** is closed. When the leadscrew **256** is driven all the way to the closed position then it can no longer rotate. The external magnet **236** will

continue to rotate and the magnets **236**, **244** will become out of phase. Eventually, the phase angle will become large enough to force an axial repulsion and activate the limit switch **240**, which will be detected, indicating that the valve **200** is completely closed. Likewise, when opening the valve **200**, activation of the limit switch **240** will occur when the valve is opened to its mechanical limit. (81) An interesting benefit to this arrangement is that pilot valves require a minimum amount of pressure to stay closed. For example, if you connect one to a water supply that is off and then turn the supply on, you will typically get a short burst of water and then the valve will seal. That does not happen with this implementation because the leadscrew **256** mechanically holds the diaphragm **220** closed.

(82) With respect to the use of alternate sensors in the pressure control valve **200**, which uses a lead screw **256**, it would make sense to keep the limit switch **240** because the hard stops are only used to detect closing the valve and opening it to its maximum travel. However, it could still benefit from using an optical sensor, though. For example, if the control chamber **212** has an optically clear window then the lead screw travel could be monitored by an encoder. That is typically implemented by including lines on the shaft that can be counted by the sensor. This allows for tracking exactly how much the screw has rotated inside the chamber.

(83) While the present pressure valve embodiment **200** is illustrated and described as being controlled by a motor assembly **202**, the valve **200** could be manually controlled or controlled by other actuators. For example, a manually controlled pilot valve assembly **200** (without motor or without motor and gears, i.e. manually rotating the external magnet itself) could find use in other applications as a conventional faucet valve or spigot (not shown).

(84) Turning now to FIGS. **16-23**, there is illustrated and described an oscillator/diverter assembly **300** which is based on a water turbine rotation mechanism wherein the position of a diverter arm is electronically controlled by a magnetic coupling mechanism **304** in accordance with the above teachings. The assembly **300** generally includes drive motor **302**, the magnetic coupling assembly **304**, a diverter assembly **306** and an oscillator drive assembly **308**.

(85) The oscillator drive portion **308** is a water powered turbine motor that rotates a shaft **310** and neck **312** to which spray nozzle **14** is secured. The rotation thereby provides the ability to direct spray in different directions. The oscillator drive portion **308** incorporates an assembly of components that allow the drive to reverse the direction by directing the flow of water in one of two orientations causing a turbine wheel **314** to rotate in one of two directions, accordingly. The turbine wheel **314** provides the rotational input force to the water powered motor, so reversing the turbine direction also reverses the motor direction. The assembly which is used to change the motor direction is, therefore, the diverter assembly **306**.

(86) The diverter assembly **306** allows the direction change to occur as a result of electrical input to the reversible electric motor **302**. The electric input is controlled by electronics, which allows the direction of the oscillator **308** to be controlled via electronic input, including microprocessor control. An additional benefit of this type of control is that the diverter **306** includes a neutral or idle position. That is, the water flow can be directed equally to both sides of the turbine wheel **314**, creating a net zero force in either direction causing the water powered motor to stop. It is also possible to control the speed of the water flow in either direction, thereby providing the ability to use the diverter **306** as a speed control for the water powered motor.

(87) FIGS. **16-18** shows various views of entire oscillator/diverter **300** for reference including a transparent elevation view.

(88) The input to the assembly, as shown, is a standard threaded garden hose connector **316**, although it could be any suitable connector or integrated directly into a common housing downstream of the pressure control valve. This is the supply pressure. Varying this pressure with the pressure control valve **200** affects the output pressure of nozzle **14** and, thus, the distance of the water sprayed from the nozzle **14**. Varying the supply pressure will also affect the speed of the water powered motor and a minimum pressure must be supplied for the motor to turn.

(89) The flow of water initially passes into a diverter chamber **318** through an input port **320** and is directed through one of two exit ports **322**, **324** in a boundary wall **326** between the diverter chamber **318** and a turbine chamber **328** in the oscillator drive portion **308**. The water flows across and rotates the turbine wheel **314** (assuming the diverter is not in the idle position). The water then flows through another boundary wall **330** into an oscillator chamber **332** containing a gear train **334**. The nozzle (not shown) is mounted to the neck **312**.

(90) There is also a pressure relief valve **338** that provides an alternative path directly from the diverter chamber **318** to the neck **312** for water at excessive supply pressures that might, otherwise, overwhelm the diverter/oscillator **300**. The pressure relief valve **338** essentially acts as a rotation speed limiter without restricting nozzle pressure. Note that restricting the nozzle pressure would restrict the maximum distance of the spray pattern.

(91) The sprinkler head gear **336** also drives a rotational sensor gear **340**, which captures a diametric permanent magnet in line with a magnetic rotational sensor **500** located external to the oscillator/diverter assembly **300**. The gear **340** has a 1:1 ratio with the head gear **336**, making it possible to electronically determine the corresponding angle of rotation of the nozzle **14**. The oscillator gear train **334**, head gear **336**, sensor gear **340** and pressure relief valve **338** are all rotatably captured within the oscillator chamber **332**, which is pressurized at the output (nozzle) pressure. The electronic pressure sensor **400** is embedded in the side wall of the oscillator chamber **332** and provides a means of electronically determining the nozzle pressure in real time. Note that the shaft **310** of neck **312** is the only component of the oscillator/diverter assembly **300** that penetrates a water chamber to atmosphere with a rotating component and, therefore, requires a seal; supplied here by an O-ring **344**. Other than the supply input **320** and the shaft **310**, the entire device is hermetically sealed. The friction introduced by the O-ring seal **344** is easily overcome by the torque produced by the gear ratio (approximately 500:1) of the oscillator drive train **334**.

(92) FIGS. **18-20** show external views of the diverter components. The motor **302** of the external portion of the magnetic coupling **304** is used to drive the mechanism. The external rotation is transferred to the internal portion of the assembly, which includes an armature **346** that diverts the water flow through one of two ports **322**, **324** into the turbine chamber **328**.

(93) The input water flow is through input port **320**. This supplies water under pressure to the diverter chamber **318**. The control input is via the external components of the magnetic coupling **304**, which changes the position of the armature **346** (FIGS. **20-23**) inside the diverter chamber **318**. The position of the armature **346** determines which port **322**, **324** the water flows through into the turbine chamber **328**. As the water flows across the turbine wheel **314** the water continues to flow through the boundary wall **330** into the oscillator chamber **332** via ports **333**. The turbine **314** rotates and provides the mechanical input to the gear train **334** via a small drive gear **348**. A relief valve port **350** further allows excess pressure to release water through to the sprinkler neck **312**, limiting the speed of the turbine **314**.

(94) FIGS. **20-21** shows the control input into the diverter armature **346**. The motor gear **352** drives the magnet gear **354**, which has a square hole that captures the external magnet **356**. The external magnet **356** is magnetically coupled to the internal magnet **358** across the boundary wall of the diverter chamber **318**. The diverter boundary wall has a round well **360** incorporated into it that captures the armature **346**. As the external magnet **356** is rotated, the internal magnet **358** is rotated accordingly. Since it is captured in a square socket in the armature **346**, the armature **346** is also rotated in correspondence with the magnetic coupling. When the magnets **356**, **358** are in phase, the north and south pole of the external magnet **356** is aligned with the south and north poles of the internal magnet **358**. In this relative position, the magnetic force vector is entirely normal to the boundary and the magnets are attracted to each other. As the motor gear **352** continues to rotate, the armature **346** contacts one of the stops **362a**, **362b**, preventing any further rotation of the armature **346**. As the external magnet **356** continues to rotate, the magnets become out of phase. When they are out of phase by 90 degrees, there is no longer a net magnetic force normal to the boundary, so

the force vector is entirely rotational. As the external magnet **356** continues to rotate putting it greater than 90 degrees out of phase with the internal magnet **358**, the magnets begin to repel each other. As described previously, the external magnet **356** is free to move normal to the boundary, so it is pushed away from the boundary and transfers the repulsive magnetic force through the magnet adapter **364** to the limit switch **366**. At some point, when the phase angle is between 90 and 180 degrees, the force becomes strong enough to activate the limit switch **366**, which is electronically detected. Upon the detection, the motor **302** is stopped and then reversed, which causes the phase angle to decrease. At some point, when the phase angle is less than 90 degrees, the magnetic force becomes attractive again and the magnet **356** moves back toward the diverter boundary, releasing the limit switch **366**. Again, the deactivation of the limit switch **366** is detected electronically, and the motor **302** is stopped and locked. At this position, the phase angle is, generally, still greater than zero. In the figure, the external magnet **356** is shown to be 45 degrees out of phase with the internal magnet. In this state, the magnetic force vector is holding the armature **346** against the diverter boundary and against the stop **362**, which is situated as to locate the armature plug **368** over one of the output ports **322**, **324** to the turbine chamber **328**.

(95) FIGS. **21-23** show the armature **346** being held in position against a stop **362**, which holds the armature plug **368** in line with one of the ports **322**, **324** (blocking the port) into the turbine chamber **328**. Blocking one port causes the water to flow into the opposite port. The pressurized water is directed onto the blades of the turbine **314** through the corresponding port outlet. In this figure, the turbine **314** would rotate counter-clockwise, as viewed from above. Rotating the armature **346** to the opposite stop would cause the plug **368** to block the opposite port, which would cause water to flow through the other port and the turbine **314** would rotate in the clockwise direction. It is this mechanism that allows the rotation direction of the oscillator to be controlled electronically. Note that it is possible to turn the motor **302** so that the armature **346** is in a position midway between the ports (position not shown). In this state, the water flows equally between the two ports **322**, **324** and the net pressure at the two port outlets is approximately the same and the turbine **314** does not rotate. One exemplary method of achieving this state is by measuring the time it takes to drive the motor **302** from one stopped position to the other and then by rotating the motor **302** from one stopped position toward the other for half of that duration. Thus, the control is able to effect three states: rotation clockwise, rotation counter-clockwise and stationary. Other possible methods include the use of artificial intelligence (AI) learning algorithms which learn and adjust motor timing. AI type learning algorithms are effective for this type of implementation because of the unpredictable nature of water in mechanical systems.

(96) Another benefit to the present smart sprinkler arrangement is that the sprinkler system will never get stuck due to low water pressure. One problem with purely mechanical sprinklers is that they require a minimum water pressure to operate. There is friction in the water motor mechanism, so you have to turn up the water pressure to some minimum amount to overcome that or it won't rotate. It can even be hard to predict because that point can be a little different along the rotation. Also, the mechanical actuator for changing direction requires an even greater pressure to overcome the force involved in moving the actuator. Moreover, the supply pressure can change, so that a drop in pressure can cause a sprinkler to stop rotating even if it was when it was set up.

(97) The present system has the same problem with friction of the water powered motor **308**. Some minimum is required for rotation of the turbine wheel **314** to occur. With the present sprinkler system **10**, the point of direction change is detected by a sensor and the change of direction is accomplished via an electric motor. Neither introduces additional friction or requires water pressure. In addition, it can detect whether the sprinkler head is rotating (rotational sensor gear **340**) and can automatically raise the pressure (valve **200**) until it starts rotating. It also raises the pressure automatically if the supply pressure drops due to outside factor (local water demand). This means that the sprinkler **10** will not get stuck. In fact, there really is no minimum pressure. The system will lower the pressure to a user set point and then automatically raise it enough to rotate

the sprinkler head **14**. Once rotation is detected, it will again lower the pressure to the user set point. The system won't generate a simple arc in this mode, as it would with a higher pressure, but it will apply water at the set range, and it won't get stuck. Note that this assumes that the supply pressure is greater than the minimum required pressure. If this is not satisfied, the sprinkler can detect the condition and shut itself off completely, abandoning the user pattern until such time as the supply pressure is restored. Similarly, if a water pattern was configured that includes a maximum water pressure that cannot be achieved at the time of running the pattern, the sprinkler can automatically skip over those portions of the pattern that cannot be achieved and still execute the portions of the pattern that can. Thus, the sprinkler can come as close to satisfying the desires of the user as is possible.

(98) Accordingly, it can be seen that the present system **10** provides several unique and novel improvements over systems of the prior art, particularly with respect to sealed chamber magnetic couplers and switching mechanism which eliminate the need for high friction seals for rotating parts and which also reduce power needs for rotating components within the sealed chambers.

(99) Location Awareness

(100) In the present system **10**, the user is given the ability to create and store patterns that the control system **22** will be able to follow in a repeatable and predictable manner. Since the two control mechanisms involve sensory awareness and control of distance (proportional to water pressure) and rotational angle, repeatability requires specific location and orientation calibration.

(101) The system **10** uses a positionable mounting base to implement this functionality. Referring to FIGS. **24-25**, there is a base **400** having a body **402** that is configured to be secured with spikes or other, semi-permanent fasteners, to the ground so as to fix the base **400** at a predetermined location. The base **400** includes a plurality of mounting feet or tabs **404** extending radially outwardly from the body **402** each with a bore **406** therethrough to accept spikes (not shown). The base **400** further incorporates a keyed mortise **408** on its top surface designed to receive a matching keyed tenon **412** on the sprinkler housing **410** to insure proper alignment of the sprinkler unit with the base **400**. In the exemplary embodiment, the mortise and tenon are each generally circular with a single flat surface for rotational orientation. Other interfitting mated alignment structures are also contemplated. With this arrangement, the sprinkler system **10** can be reliably placed on the base **400** in a specific location and rotational orientation. If the base is not moved, any patterns created with the sprinkler on the base can be run repeatedly and reproduce the same result.

(102) Moreover, any number of a plurality of bases such as **400**, **400'** and **400''** may be located on the ground to provide a plurality of defined coverage areas, and repeatability of each of the pre-programmed coverage patterns for the respective coverage areas will be maintained, so long as the pattern being run was created on the particular base and is selected for operation on the associated base. In this way, a plurality of bases **400** allows for a corresponding expansion of the useful geographic range of the sprinkler system **10**.

(103) In order to provide automated "awareness" of location and selection of a predetermined sprinkler pattern for that location, each base incorporates an RFID tag **414** (See FIGS. **24-25**), which is a passive electronic component requiring no electrical power in its idle state. The lower portion of the sprinkler housing **410** incorporates an RFID reader (antenna TX/RX) **416** that is aligned with the RFID tag **414** when the sprinkler housing **410** is placed on any given base (**400**, **400'**, **400''**). A corresponding antenna within the RFID tag allows a transmitted signal from a nearby electronic circuit (reader **416**) to energize the passive RFID tag **414** and to read its unique identifier. Once the identifier is known, the system software can automatically relate the identifier to a known base and, therefore, to a known location and an associated pattern, or collection of patterns, saved in memory. This alleviates the burden on the user of manually associating the pre-programmed patterns with the individual bases.

(104) It should also be noted that the system **10** can also detect the condition where the sprinkler is not on any base and can, therefore, detect the eventuality of the sprinkler being placed on or taken

off a base. This allows an associated pattern to be automatically initiated when the sprinkler is placed on a particular base. Conversely, a running pattern may be interrupted when the sprinkler is removed from a base and resumed when placed back on the same base.

(105) An analogous ID system could be implemented using barcodes or other unique identifier schemes, but the RFID tag is much more robust for this particular use scenario.

(106) Gesture Support

(107) The exemplary embodiment of the present sprinkler system **10** has no mechanical features on the physical device for manually controlling any aspect of the water spray. Rather, the system **10** is designed to use a separate device **20** as a user interface with all control input being transmitted from the user interface device **20** to the sprinkler **10** over a communication interface.

(108) In some embodiments, in order to provide a manual input scheme that can be used without the communication interface **34**, the sprinkler may include an accelerometer **500** (2 axis or 3 axis) to facilitate certain device functions without the need for the user interface device (see FIG. **1** block diagram scheme).

(109) Accelerometer **500** is a well-known electronic component that may be affixed to the internal structure of the housing **410** or directly mounted to an internal circuit board. The accelerometer **500** allows instantaneous detection of the orientation of the sprinkler housing **410**. Software algorithms are used to interpret real time changes in orientation as “tilt” gestures. Various tilt gestures can be distinguished for the purpose of interpreting user intent and thus provide a reliable manual input regarding desired function without externally facing switches.

(110) Some examples follow:

(111) If the sprinkler is off, then a rocking or rotating motion of the housing **410** while the sprinkler is not on a base (see RFID above) then it may be assumed that the user wants to turn on the water for use in a manual mode (i.e., not running a pattern). The control would then open the pilot valve to a predetermined position. A similar or same rocking motion can be interpreted as user intent to turn the water back off.

(112) If the water is on in this manual mode, then turning the sprinkler sideways could indicate a desire to turn off the water temporarily. For instance, the user may lay the sprinkler on its side on the ground. Returning the sprinkler to its upright position could then be interpreted as resuming water flow.

(113) Also while the water is on in manual mode, a sudden impact, i.e. tapping or bumping the sprinkler housing against an opposing hand could be interpreted to increase water flow from a lower to a higher flow or vice versa.

(114) Other combinations of orientation changes, insofar as they are identifiable, could be mapped to other functionality, such as rotating the sprinkler nozzle to a different rotational angle.

(115) Having thus described certain particular embodiments of the invention, it is understood that the invention defined by the appended claims is not to be limited by particular details set forth in the above description, as many apparent variations thereof are contemplated. Rather, the invention is limited only by the appended claims, which include within their scope all equivalent devices or methods which operate according to the principles of the invention as described.

Claims

1. A sprinkler system comprising: a housing; a pressure control valve associated with a first control motor; a second control motor; a spray head controlled by said second control motor for oscillating rotation thereof; a motor controller connected to said first control motor and said second control motor, said motor controller including a processor and a memory; a programming device in selective communication with said motor controller, said programming device running a software application which is programmed to: receive input by an interactive user interface to define a predefined watering pattern; communicate said predefined watering pattern to said motor

controller; store said predefined watering pattern in said motor controller; and initiate a run sequence for automated operation of said first control motor and said second control motor to complete said predefined watering pattern; a base, said housing and said base including interfitting mating formations for rotational alignment of the housing relative to the base; an RFID tag on the base, said RFID tag including a unique identifier, said unique identifier being associated with said predefined watering pattern saved in said memory; and an RFID reader within the housing and in communication with the motor controller processor, wherein placement of the housing on the base automatically associates the predefined watering pattern with the base.

2. The sprinkler system of claim 1 further comprising a plurality of bases each including an RFID tag with a unique identifier, and a plurality of predefined watering patterns saved in said memory, each of said plurality of predefined watering patterns being associated with a respective one of said plurality of bases.

3. The sprinkler system of claim 1 wherein the base includes a plurality of mounting feet extending radially outwardly from the body, each foot including a bore therethrough.

4. The sprinkler system of claim 2 wherein each base includes a plurality of mounting feet extending radially outwardly from the body, each foot including a bore therethrough.

5. The sprinkler system of claim 1 wherein the interfitting mating formations comprise a keyed mortise on a top surface of a base and a corresponding keyed tenon on the sprinkler housing.

6. The sprinkler system of claim 2 wherein the interfitting mating formations comprise a keyed mortise on a top surface of each base and a corresponding keyed tenon on the sprinkler housing.

7. The sprinkler system of claim 3 wherein the interfitting mating formations comprise a keyed mortise on a top surface of a base and a corresponding keyed tenon on the sprinkler housing.

8. The sprinkler system of claim 4 wherein the interfitting mating formations comprise a keyed mortise on a top surface of each base and a corresponding keyed tenon on the sprinkler housing.

9. The sprinkler system of claim 1 further comprising an accelerometer within said housing in communication with said motor controller, wherein predefined movements of said housing are interpreted as inputs to said motor controller for selectively controlling operation of said motor controller.

10. The sprinkler system of claim 1 further comprising an external sensor in communication with said motor controller to provide additional operational input.

11. A sprinkler system comprising: a housing; a pressure control valve associated with a first control motor; a second control motor; a spray head controlled by said second control motor for oscillating rotation thereof; a motor controller connected to said first control motor and said second control motor, said motor controller including a processor and a memory; a programming device in selective communication with said motor controller, said programming device running a software application which is programmed to: receive input by an interactive user interface to define a predefined watering pattern; communicate said predetermined watering pattern to said motor controller; store said predefined watering pattern in said motor controller; and initiate a run sequence for automated operation of said first control motor and said second control motor to complete said watering pattern; and an accelerometer within said housing in communication with said motor controller, wherein predefined manual gesture movements of said housing by a user are interpreted as inputs to said motor controller for selectively controlling operation of said motor controller.

12. The sprinkler system of claim 11, further comprising: a base, said housing and said base including interfitting mating formations for rotational alignment of the housing relative to the base.

13. The sprinkler system of claim 12 further comprising: an RFID tag on the base, said RFID tag including a unique identifier, said unique identifier being associated with said predefined watering pattern saved in said memory; and an RFID reader within the housing and in communication with the motor controller processor, wherein placement of the housing on the base automatically associates the predefined watering pattern with the base.

14. The sprinkler system of claim 12 wherein the interfitting mating formations comprise a keyed mortise on a top surface of a base and a corresponding keyed tenon on the sprinkler housing.
 15. The sprinkler system of claim 12 wherein the base includes a plurality of mounting feet extending radially outwardly from the body, each foot including a bore therethrough.
 16. The sprinkler system of claim 12 further comprising a plurality of bases each including an RFID tag with a unique identifier, and a plurality of predefined watering patterns saved in said member, each of said plurality of predefined watering patterns being associated with a respective one of said plurality of bases.
 17. The sprinkler system of claim 16 wherein each base includes a plurality of mounting feet extending radially outwardly from the body, each foot including a bore therethrough.
 18. The sprinkler system of claim 10 wherein said external sensor is selected from the group consisting of: a temperature sensor, a light sensor, a rain sensor, a pressure sensor, and an infra-red heat sensor.
 19. The sprinkler system of claim 11 further comprising an external sensor in communication with said motor controller to provide additional operational input.
 20. The sprinkler system of claim 19 wherein said external sensor is selected from the group consisting of: a temperature sensor, a light sensor, a rain sensor, a pressure sensor, and an infra-red heat sensor.
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